Topic B4: Community level systems

B4.1 Ecosystems

Summary

Microorganisms play an important role in the continuous cycling of chemicals in ecosystems. Biotic and abiotic factors interact in an ecosystem and have an effect on communities. Living organisms form populations of single species, communities of many species and are part of ecosystems. Living organisms are interdependent and show adaptations to their environment. Feeding relationships reflect the stability of an ecosystem and indicate the flow of biomass through the ecosystem.

Underlying knowledge and understanding

Learners should be familiar with the idea of a food web and the interrelationships associated with them and that variation allows living things to survive in the same ecosystem. They should also recognise that organisms affect their environment and are affected by it.

Common misconceptions

Research has shown that it is easier for a learner to explain the consequences on a food web if the producers are removed for some reason than if the top predators are taken away. It is also better to start off explaining ideas relating to food webs using small simple webs with animals and plants that learners are likely to know e.g. rabbits and foxes. Learners find arrows showing the flow of biomass from one trophic level to another quite challenging and often mistake it for the direction of predation. This makes problems relating to the manipulation of a food web quite difficult for some.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM4.1i ☑	calculate rate changes in the decay of biological material	M1c
BM4.1ii	calculate the percentage of mass	M1c
BM4.1iii ☑	Use fractions and percentages	M1c
BM4.1iv	plot and draw appropriate graphs selecting appropriate scales for the axes	M4a and M4c
BM4.1v ☑	extract and interpret information from charts, graphs and tables	M2c and M4a

B4.1a	recall that many different materials cycle through the abiotic and biotic components of an ecosystem	examples of cycled materials e.g. nitrogen and carbon			
B4.1b	explain the role of microorganisms in the cycling of materials through an ecosystem	the role of microorganisms in decomposition			Research into the range of ecosystems and examples of micro-organisms that act as decomposers within them. (PAG B1, PAG B3, PAG B4, PAG B7)
B4.1c	explain the importance of the carbon cycle and the water cycle to living organisms	maintaining habitats, fresh water, flow of nutrients and the stages of the carbon and water cycles			
B4.1d ☑	explain the effect of factors such as temperature, water content, and oxygen availability on rate of decomposition	the terms aerobic and anaerobic	M1c, M2c, M4a, M4c	WS1.1b, WS1.1h, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3f, WS1.3g, WS2a, WS2b, WS2c, WS2d	Investigation of the most favourable conditions for composting. (PAG B1, PAG B3, PAG B4, PAG B7)
B4.1e	describe different levels of organisation in an ecosystem from individual organisms to the whole ecosystem		M1c		
B4.1f	explain how abiotic and biotic factors can affect communities	temperature, light intensity, moisture level, pH of soil, predators, food	M3a, M4a, M4c	WS1.3a, WS1.3b, WS1.3e WS1.3h, WS2a, WS2b, WS2c, WS2d	Identification of the biotic factors in an ecosystem using sampling techniques. (PAG B3)

B4.1g	describe the importance of interdependence and competition in a community	interdependence relating to predation, mutualism and parasitism		WS1.4a, WS2a, WS2b, WS2c, WS2d	Examination of the roots of a leguminous plant e.g. clover to observe the root nodules. (PAG B1) Investigation of the holly leaf miner or the horse-chestnut leaf miner (*Cameraria ohridella*) (PAG B1, PAG B3)
B4.1h ☑	describe the differences between the trophic levels of organisms within an ecosystem	use of the terms producer and consumer			Investigation of the trophic levels within a children's story (e.g. *The Gruffalo*)
B4.1i ☑	describe pyramids of biomass and explain, with examples, how biomass is lost between the different trophic levels	loss of biomass related to egestion, excretion, respiration	M1c, M4a	WS1.3c, WS1.3e	Discussion of the best food source for humans (e.g. 'wheat vs. meat') Production of ecological pyramids.
B4.1j ☑	calculate the efficiency of biomass transfers between trophic levels and explain how this affects the number of trophic levels in a food chain		M1c		Calculation of the biomass transfers using real data.